

MHS PACKAGE: EQUIVARIANT SMOOTH HYPERCOVERS AND COHOMOLOGY

Let $f : X \rightarrow X$ be an automorphism of a projective variety. We work over $\mathbb{C}$, and X may be singular and nonnormal.

Step 0: the space X_0 . Let $\pi_0 : X_0 \rightarrow X$ be an f -equivariant projective resolution supplied by functorial resolution. Thus X_0 is smooth projective and f lifts to an automorphism $f_0 : X_0 \rightarrow X_0$ satisfying

$$\pi_0 \circ f_0 = f \circ \pi_0.$$

Such a resolution exists in characteristic zero [BM08, Theorem 1.1]. Put $U_0 = X_0$, $g_0 = f_0$, and $\eta_0 = \pi_0$. At every subsequent step we use the same type of functorial resolution for reduced projective schemes, allowing a finite disjoint union of smooth projective components as the source.

Step 1: the space X_1 , its two faces, and its degeneracy. Form

$$Y_1 = X_0 \times_X X_0, \quad h_1 = f_0 \times_X f_0.$$

Let $\tilde{\sigma}_1 : U_1 \rightarrow (Y_1)_{\text{red}}$ be an h_1 -equivariant projective resolution, with lift $g_1 \in \text{Aut}(U_1)$. Put

$$\iota_1 : (Y_1)_{\text{red}} \hookrightarrow Y_1, \quad \sigma_1 = \iota_1 \tilde{\sigma}_1, \quad \sigma_1 g_1 = h_1 \sigma_1.$$

Number the coordinates of Y_1 by 0, 1, and set

$$b_0^{(1)} = \text{pr}_1 \sigma_1, \quad b_1^{(1)} = \text{pr}_0 \sigma_1, \quad \eta_1 = \pi_0 b_0^{(1)} = \pi_0 b_1^{(1)}.$$

The index of a face is the omitted vertex: on a pair (z_0, z_1) , d_0 retains z_1 and d_1 retains z_0 . Define

$$X_1 = U_1 \sqcup U_0.$$

Write $j_1 : U_1 \hookrightarrow X_1$ for the first inclusion and $s_0^{(0)} : X_0 = U_0 \hookrightarrow X_1$ for the second. Define $d_0^{(1)}, d_1^{(1)} : X_1 \rightarrow X_0$ by their restrictions:

$$(1) \quad d_i^{(1)} j_1 = b_i^{(1)}, \quad d_i^{(1)} s_0^{(0)} = \text{id}_{X_0} \quad (i = 0, 1).$$

The augmentation and automorphism are

$$\pi_1|_{U_1} = \eta_1, \quad \pi_1|_{U_0} = \pi_0, \quad f_1 = g_1 \sqcup g_0.$$

In particular, $(d_1^{(1)}, d_0^{(1)}) : X_1 \rightarrow Y_1$ restricts to σ_1 on U_1 and to the diagonal on U_0 , so it is projective and surjective. The extra U_0 component supplies the degeneracy $s_0^{(0)}$; a lift of the diagonal to U_1 is not required.

Step 2: the space X_2 and all its structure maps. For the next matching object it is convenient to index coordinates by faces. Let Y_2 be the closed subscheme of $X_1 \times_X X_1$ given by

$$(2) \quad Y_2 = \left\{ (z_0, z_1, z_2) \left| \begin{array}{l} d_0^{(1)} z_1 = d_0^{(1)} z_0, \\ d_0^{(1)} z_2 = d_1^{(1)} z_0, \\ d_1^{(1)} z_2 = d_1^{(1)} z_1 \end{array} \right. \right\}.$$

These are scheme-theoretic equalizers. Put $h_2 = (f_1 \times_X f_1 \times_X f_1)|_{Y_2} \in \text{Aut}(Y_2)$.

Let $\tilde{\sigma}_2 : U_2 \rightarrow (Y_2)_{\text{red}}$ be an h_2 -equivariant projective resolution, with lift $g_2 \in \text{Aut}(U_2)$. Put

$$\iota_2 : (Y_2)_{\text{red}} \hookrightarrow Y_2, \quad \sigma_2 = \iota_2 \tilde{\sigma}_2, \quad \sigma_2 g_2 = h_2 \sigma_2, \\ b_i^{(2)} = \text{pr}_i \sigma_2 \quad (0 \leq i \leq 2).$$

Set

$$X_2 = U_2 \sqcup U_1^{(0)} \sqcup U_1^{(1)} \sqcup U_0^{(00)},$$

where superscripts label disjoint copies. Denote the four inclusions by $j_2, j_{10}, j_{11}, j_{00}$, respectively. The three face maps $d_i^{(2)} : X_2 \rightarrow X_1$ are exactly the following maps on these copies:

component	$d_0^{(2)}$	$d_1^{(2)}$	$d_2^{(2)}$
U_2	$b_0^{(2)}$	$b_1^{(2)}$	$b_2^{(2)}$
$U_1^{(0)}$	j_1	j_1	$s_0^{(0)} b_1^{(1)}$
$U_1^{(1)}$	$s_0^{(0)} b_0^{(1)}$	j_1	j_1
$U_0^{(00)}$	$s_0^{(0)}$	$s_0^{(0)}$	$s_0^{(0)}$

Define the two degeneracies $s_0^{(1)}, s_1^{(1)} : X_1 \rightarrow X_2$ by

$$(3) \quad \begin{aligned} s_0^{(1)} j_1 &= j_{10}, & s_1^{(1)} j_1 &= j_{11}, \\ s_0^{(1)} s_0^{(0)} &= s_1^{(1)} s_0^{(0)} = j_{00}. \end{aligned}$$

Finally put

$$\pi_2 = \pi_1 d_0^{(2)} = \pi_1 d_1^{(2)} = \pi_1 d_2^{(2)}, \quad f_2 = g_2 \sqcup g_1 \sqcup g_1 \sqcup g_0.$$

The equations defining Y_2 give the face identities on U_2 , and the table verifies them on the other components. It also verifies all face–degeneracy identities in these degrees.

Step p: the general induction. Suppose that the levels below p have been constructed, where $p \geq 3$. Define the matching scheme

$$(4) \quad Y_p = \left\{ (z_0, \dots, z_p) \in \underbrace{X_{p-1} \times_X \cdots \times_X X_{p-1}}_{p+1 \text{ factors}} \mid d_i^{(p-1)} z_j = d_{j-1}^{(p-1)} z_i \ (i < j) \right\}.$$

As in (2), this is a scheme-theoretic equalizer. It is projective because X_{p-1} is projective and the equalizers are closed. The componentwise action of f_{p-1} preserves the equations, giving $h_p \in \text{Aut}(Y_p)$. Let $\tilde{\sigma}_p : U_p \rightarrow (Y_p)_{\text{red}}$ be an h_p -equivariant projective resolution, with lift $g_p \in \text{Aut}(U_p)$. Let $\iota_p : (Y_p)_{\text{red}} \hookrightarrow Y_p$ be the closed immersion. Adjoin the degeneracy copies of the earlier pieces to form

$$X_p = U_p \sqcup X_p^{\text{deg}}, \quad X_p^{\text{deg}} := \coprod_{0 \leq q < p} \coprod_{\sigma : [p] \twoheadrightarrow [q]} U_q^{\sigma},$$

where the surjections are order-preserving and U_q^{σ} is a labelled copy of U_q . Let $j_p : U_p \hookrightarrow X_p$ be the inclusion. The automorphism f_p acts as g_p on U_p and as g_q on each copy U_q^{σ} , leaving its label fixed. The construction is summarized below; the maps on the degeneracy copies are specified in (10)–(11).

$$(5) \quad \begin{array}{ccccc} \begin{array}{c} g_p \\ \circlearrowleft \\ U_p \end{array} & \xrightarrow{\tilde{\sigma}_p} & \begin{array}{c} (h_p)_{\text{red}} \\ \circlearrowleft \\ (Y_p)_{\text{red}} \end{array} & \xrightarrow{\iota_p} & \begin{array}{c} h_p \\ \circlearrowleft \\ Y_p \end{array} \\ \downarrow j_p & & \nearrow \tau_p & & \downarrow \text{pr}_i \\ X_p = U_p \sqcup X_p^{\text{deg}} & \xrightarrow{d_i^{(p)}} & X_{p-1} & \xrightarrow{\pi_{p-1}} & X. \\ \begin{array}{c} \circlearrowleft \\ f_p \end{array} & & \begin{array}{c} \circlearrowleft \\ f_{p-1} \end{array} & & \begin{array}{c} \circlearrowleft \\ f \end{array} \end{array}$$

Each loop denotes the indicated automorphism; every arrow intertwines the automorphisms at its source and target. Here $\tau_p = (d_0^{(p)}, \dots, d_p^{(p)}) : X_p \rightarrow Y_p$ is the matching map, and, for $0 \leq i \leq p$,

$$\sigma_p := \iota_p \tilde{\sigma}_p = \tau_p j_p, \quad b_i^{(p)} := \text{pr}_i \sigma_p = d_i^{(p)} j_p, \quad \eta_p := \pi_{p-1} b_i^{(p)} = \pi_p j_p.$$

In particular, the new piece U_p maps surjectively onto Y_p , whereas X_p^{deg} supplies the degeneracies. Since $Y_p \subseteq X_{p-1} \times_X \dots \times_X X_{p-1}$, η_p is independent of i . For $0 \leq i < j \leq p$, the matching equations give the commutative square

$$(6) \quad \begin{array}{ccc} U_p & \xrightarrow{b_j^{(p)}} & X_{p-1} \\ b_i^{(p)} \downarrow & & \downarrow d_i^{(p-1)} \\ X_{p-1} & \xrightarrow{d_{j-1}^{(p-1)}} & X_{p-2}. \end{array}$$

For $p = 2$, the same diagrams define η_2 and express the identities following from (2).

Definition of the index set and the copies U_q^σ . For $0 \leq q \leq p$, put $[p] = \{0, \dots, p\}$ and

$$S_{p,q} = \{\sigma : [p] \rightarrow [q] \mid \sigma \text{ is order-preserving and surjective}\}.$$

Equivalently, a label $\sigma \in S_{p,q}$ is the tuple $(\sigma(0), \dots, \sigma(p))$ satisfying

$$\sigma(0) = 0, \quad \sigma(p) = q, \quad \sigma(i+1) - \sigma(i) \in \{0, 1\} \quad (0 \leq i < p).$$

Thus each fibre $\sigma^{-1}(j)$ is a nonempty consecutive block. There are $\binom{p}{q}$ labels: choose the q positions, among the p adjacent pairs, at which the value increases.

The scheme U_q was constructed at degree q . For each $\sigma \in S_{p,q}$, define a labelled copy

$$U_q^\sigma := U_q \times_{\mathbb{C}} \{\sigma\}, \quad \{\sigma\} := \text{Spec } \mathbb{C} \text{ with label } \sigma.$$

The projection $U_q^\sigma \rightarrow U_q$ is an isomorphism. Define

$$(7) \quad X_p := \coprod_{q=0}^p \coprod_{\sigma \in S_{p,q}} U_q^\sigma, \quad j_\sigma^{(p)} : U_q \xrightarrow{\sim} U_q^\sigma \hookrightarrow X_p.$$

The last map is an open-and-closed immersion. In labelled-point notation,

$$j_\sigma^{(p)}(u) = (u, \sigma), \quad U_q^\sigma \cap U_{q'}^{\sigma'} = \emptyset \quad \text{if } (q, \sigma) \neq (q', \sigma').$$

Here the superscript records which copy occurs in X_p ; it does not change the scheme U_q . The identity in $S_{p,p}$ gives the new piece U_p . The terms with $q < p$ are the degeneracy copies of earlier pieces. Each U_q itself may have several connected components.

For $p = 1$, the complete list is

$$S_{1,1} = \{(0, 1)\}, \quad S_{1,0} = \{(0, 0)\}, \quad X_1 = U_1^{(0,1)} \sqcup U_0^{(0,0)},$$

$$j_{(0,1)}^{(1)} = j_1, \quad j_{(0,0)}^{(1)} = s_0^{(0)}.$$

For $p = 2$, it is

$$S_{2,2} = \{(0, 1, 2)\}, \quad S_{2,1} = \{(0, 0, 1), (0, 1, 1)\}, \quad S_{2,0} = \{(0, 0, 0)\},$$

$$X_2 = U_2^{(0,1,2)} \sqcup U_1^{(0,0,1)} \sqcup U_1^{(0,1,1)} \sqcup U_0^{(0,0,0)}.$$

The labels and the notation of Step 2 correspond as follows:

σ	copy in Step 2	$j_\sigma^{(2)}$
$(0, 1, 2)$	U_2	j_2
$(0, 0, 1)$	$U_1^{(0)}$	$s_0^{(1)} j_1 = j_{10}$
$(0, 1, 1)$	$U_1^{(1)}$	$s_1^{(1)} j_1 = j_{11}$
$(0, 0, 0)$	$U_0^{(00)}$	$s_0^{(1)} s_0^{(0)} = s_1^{(1)} s_0^{(0)} = j_{00}$

In particular, the two expressions in the last row give the same copy of U_0 . The indexing records the composite surjection, so that two expressions for the same degeneracy are counted only once.

For every $\sigma \in S_{p,q}$ define the augmentation and automorphism by

$$(8) \quad \pi_p j_\sigma^{(p)} = \eta_q, \quad f_p j_\sigma^{(p)} = j_\sigma^{(p)} g_q.$$

Equivalently, $\pi_p(u, \sigma) = \eta_q(u)$ and $f_p(u, \sigma) = (g_q(u), \sigma)$.

Here are formulas for every face and degeneracy. Let $\partial_i : [p-1] \hookrightarrow [p]$ omit i , and let $\epsilon_i : [p] \twoheadrightarrow [p-1]$ identify i and $i+1$. Explicitly,

$$\partial_i(k) = \begin{cases} k & k < i, \\ k+1 & k \geq i, \end{cases} \quad \epsilon_i(k) = \begin{cases} k & k \leq i, \\ k-1 & k > i. \end{cases}$$

For any order-preserving surjection $\tau : [m] \twoheadrightarrow [r]$, let $J_\tau : X_r \rightarrow X_m$ be the map that only relabels copies:

$$(9) \quad J_\tau j_\rho^{(r)} = j_{\rho\tau}^{(m)} \quad (\rho : [r] \twoheadrightarrow [t]).$$

On the copy $U_t^\rho \subset X_r$, this is the isomorphism onto $U_t^{\rho\tau} \subset X_m$ given by

$$J_\tau(u, \rho) = (u, \rho \circ \tau).$$

For surjections $\tau : [m] \twoheadrightarrow [r]$ and $\kappa : [n] \twoheadrightarrow [m]$, the definitions give

$$J_{\text{id}_{[r]}} = \text{id}_{X_r}, \quad J_\kappa \circ J_\tau = J_{\tau \circ \kappa}.$$

Thus J_τ is defined directly from the disjoint unions, before checking any simplicial identities.

For a component indexed by $\sigma : [p] \twoheadrightarrow [q]$, the map $\sigma \partial_i$ is either surjective, or misses exactly one value $j \in [q]$. In the latter case it factors uniquely as $\sigma \partial_i = \partial_j \tau$ with $\tau : [p-1] \twoheadrightarrow [q-1]$. Define

$$(10) \quad d_i^{(p)} j_\sigma^{(p)} = \begin{cases} j_{\sigma \partial_i}^{(p-1)}, & \sigma \partial_i \text{ is surjective,} \\ J_\tau b_j^{(q)}, & \sigma \partial_i = \partial_j \tau. \end{cases}$$

The second case has $q \geq 1$, so $b_j^{(q)} : U_q \rightarrow X_{q-1}$ is already defined (including (5) when $q = p$). In particular, $d_i^{(p)}|_{U_p} = b_i^{(p)}$. Define the incoming degeneracies by

$$(11) \quad s_i^{(p-1)} j_\sigma^{(p-1)} = j_{\sigma \epsilon_i}^{(p)} \quad (\sigma : [p-1] \twoheadrightarrow [q], 0 \leq i \leq p-1).$$

Equivalently, $s_i^{(p-1)} = J_{\epsilon_i}$, and on labelled copies

$$s_i^{(p-1)}(u, \sigma) = (u, \sigma \circ \epsilon_i).$$

The label repeats its i -th entry; the point of U_q is unchanged. These rules agree with the formulas in Steps 1 and 2.

Verification of the simplicial identities and equivariance. The maps just defined satisfy

$$(12) \quad d_i^{(p-1)} d_j^{(p)} = d_{j-1}^{(p-1)} d_i^{(p)} \quad (0 \leq i < j \leq p),$$

$$(13) \quad s_i^{(p+1)} s_j^{(p)} = s_{j+1}^{(p+1)} s_i^{(p)} \quad (0 \leq i \leq j \leq p),$$

where the first identity is used for $p \geq 2$. For $p \geq 0$, $0 \leq i \leq p+1$, and $0 \leq j \leq p$, the remaining identities are

$$(14) \quad d_i^{(p+1)} s_j^{(p)} = \begin{cases} s_{j-1}^{(p-1)} d_i^{(p)}, & i < j, \\ \text{id}_{X_p}, & i = j \text{ or } i = j+1, \\ s_j^{(p-1)} d_{i-1}^{(p)}, & i > j+1. \end{cases}$$

To check these, restrict to a component U_q^σ . For (13), both sides relabel it by the same composite surjection. For (14), deleting a repeated index either cancels the repetition or moves past it; the three cases are precisely (10) applied to $\sigma \epsilon_j \partial_i$. For (12), two deleted indices either give the same relabeling, or require boundary maps $b_j^{(q)}$. Two successive boundary maps agree by (6), while identities on lower levels hold by induction. Equivalently, one uses the unique factorization of an order-preserving map as a surjection followed by an injection. This verifies the identities as morphisms on each disjoint component. It is the explicit split extension of [D74, Proposition (6.2.4)].

The same componentwise formulas and $b_i^{(q)} g_q = f_{q-1} b_i^{(q)}$ give

$$(15) \quad \begin{aligned} \pi_p f_p &= f \pi_p, & d_i^{(p)} f_p &= f_{p-1} d_i^{(p)}, & s_i^{(p)} f_p &= f_{p+1} s_i^{(p)}, \\ \pi_{p-1} d_i^{(p)} &= \pi_p, & \pi_{p+1} s_i^{(p)} &= \pi_p, \end{aligned}$$

whenever the indices are defined. The matching map $X_p \rightarrow Y_p$ restricts to σ_p on U_p (with the coordinate convention of Step 1 for $p = 1$), so it is projective and surjective. Every X_p is a finite disjoint union of smooth projective varieties; the copy of U_0 makes $\pi_p : X_p \rightarrow X$ surjective.

These are exactly the conditions for a *split proper hypercover* $\pi_\bullet : X_\bullet \rightarrow X$. In the usual notation,

$$Y_p \simeq \left(\text{cosk}_{p-1}^X \text{sk}_{p-1} X_\bullet \right)_p.$$

Here sk_{p-1} retains levels $0, \dots, p-1$ and their structure maps, and cosk_{p-1}^X is its right adjoint in schemes over X . The equalizer formula (4) is the concrete meaning of this matching object for $p \geq 2$ [Stacks, Tag 0AMA, Lemma 14.19.5].

Theorem 1. *For the system $(X_\bullet, \pi_\bullet, f_\bullet)$ constructed above, the following assertions hold. There are also Cartier divisors D_p on X_p satisfying part (4) below.*

- (1) *The maps π_p are projective and surjective, every X_p is a finite disjoint union of smooth projective varieties, and the face, degeneracy, and automorphism maps satisfy (12)–(15).*
- (2) *Write $H^k(Z, \mathbb{Q}) = H^k(Z^{\text{an}}, \mathbb{Q})$ for ordinary singular cohomology. The augmentation computes cohomology:*

$$(16) \quad \pi_\bullet^* : H^k(X, \mathbb{Q}) \xrightarrow{\sim} H^k(X_\bullet, \mathbb{Q}) \quad (k \geq 0).$$

Here $H^k(X_\bullet, \mathbb{Q})$ is the cohomology of the total complex of singular cochains on the levels X_p . In particular, there is a convergent spectral sequence

$$(17) \quad E_1^{p,q} = H^q(X_p, \mathbb{Q}) \implies H^{p+q}(X, \mathbb{Q}), \quad p, q \geq 0,$$

whose first differential is

$$(18) \quad \delta^{p,q} := d_1^{p,q} = \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^* : H^q(X_p, \mathbb{Q}) \longrightarrow H^q(X_{p+1}, \mathbb{Q}).$$

The spectral sequence is a spectral sequence of rational mixed Hodge structures. The operators f_p^ act on its first page, commute with its differentials, and induce f^* on the abutment.*

- (3) *Each $E_1^{p,q}$ is pure of weight q , and the spectral sequence degenerates at E_2 . For all $k \geq 0$ and $0 \leq w \leq k$, there are f^* -equivariant isomorphisms of pure Hodge structures*

$$(19) \quad \text{Gr}_w^W H^k(X, \mathbb{Q}) \simeq E_2^{k-w,w} = \frac{\ker \delta^{k-w,w}}{\text{im } \delta^{k-w-1,w}},$$

where an incoming term with negative simplicial degree is zero. Here W is Deligne's weight filtration, $\text{Gr}_w^W H = W_w H / W_{w-1} H$, and

$$W_{-1} H^k(X, \mathbb{Q}) = 0, \quad W_k H^k(X, \mathbb{Q}) = H^k(X, \mathbb{Q}).$$

The pullback to the zeroth level satisfies

$$(20) \quad \ker(\pi_0^* : H^k(X, \mathbb{Q}) \rightarrow H^k(X_0, \mathbb{Q})) = W_{k-1} H^k(X, \mathbb{Q}).$$

- (4) *The divisors satisfy*

$$(21) \quad f_p^* D_p \sim D_p, \quad D_p \text{ is } \pi_p\text{-ample.}$$

Equivalently, $M_p := \mathcal{O}_{X_p}(D_p)$ satisfies

$$(22) \quad f_p^* M_p \simeq M_p, \quad M_p \text{ is } \pi_p\text{-ample.}$$

For any ample Cartier divisor H on X and every fixed p , the class

$$(23) \quad [D_p] + m \pi_p^*[H]$$

is ample on X_p for all sufficiently large integers m .

What this gives in degree two. Formula (19) reads explicitly

$$(24) \quad \mathrm{Gr}_2^W H^2(X, \mathbb{Q}) \simeq \ker\left(H^2(X_0, \mathbb{Q}) \xrightarrow{\delta^{0,2}} H^2(X_1, \mathbb{Q})\right),$$

$$(25) \quad \mathrm{Gr}_1^W H^2(X, \mathbb{Q}) \simeq \frac{\ker\left(H^1(X_1, \mathbb{Q}) \xrightarrow{\delta^{1,1}} H^1(X_2, \mathbb{Q})\right)}{\mathrm{im}\left(H^1(X_0, \mathbb{Q}) \xrightarrow{\delta^{0,1}} H^1(X_1, \mathbb{Q})\right)},$$

$$(26) \quad \mathrm{Gr}_0^W H^2(X, \mathbb{Q}) \simeq \frac{\ker\left(H^0(X_2, \mathbb{Q}) \xrightarrow{\delta^{2,0}} H^0(X_3, \mathbb{Q})\right)}{\mathrm{im}\left(H^0(X_1, \mathbb{Q}) \xrightarrow{\delta^{1,0}} H^0(X_2, \mathbb{Q})\right)}.$$

Thus the higher levels recover the lower-weight information lost under π_0^* . These formulas compute the graded pieces; they do not assert a canonical direct-sum decomposition of the mixed Hodge structure. In particular, if f_0^* on $H^2(X_0, \mathbb{Q})$ and f_1^* on $H^1(X_1, \mathbb{Q})$ are semisimple, then f^* on $H^2(X, \mathbb{Q})$ is semisimple. The weight-zero term needs no additional hypothesis, since f_2^* permutes the finitely many connected components of X_2 .

Proof. The geometric assertions and equivariance have been proved in the construction. We construct the divisors in part (4) and then establish the cohomological assertions.

Invariant relatively ample divisors. For a reduced projective scheme T with automorphism h , a functorial resolution is a sequence of invariant-centre blowups

$$T = T_0 \xleftarrow{\sigma_1} T_1 \xleftarrow{\sigma_2} \cdots \xleftarrow{\sigma_s} T_s = \tilde{T}.$$

If E_j is the exceptional Cartier divisor of σ_j , then $\mathcal{O}_{T_j}(-E_j) \simeq \mathcal{O}_{T_j}(1)$ is σ_j -ample and invariant. Starting with $F_0 = 0$, the choice

$$(27) \quad F_j = -E_j + N_j \sigma_j^* F_{j-1}, \quad N_j \gg 0,$$

gives an invariant Cartier divisor ample relative to $T_j \rightarrow T$. Here we use functorial resolution [BM08, Theorem 1.1], the blowup identities [Stacks, Tag 02OS, Tag 02NS, Tag 0809], and the composition lemma for relative ampleness [Stacks, Tag 0C4K].

Apply this first to $U_0 \rightarrow X$, obtaining an invariant relatively ample line bundle L_0 on U_0 . Set $M_0 = L_0$. Suppose the bundles have been chosen below level p . On $T_p = (Y_p)_{\mathrm{red}}$, let $\rho_i : T_p \rightarrow X_{p-1}$ be the coordinate projections. The line bundle

$$B_p = \bigotimes_{i=0}^p \rho_i^* M_{p-1}$$

is h_p -invariant and ample relative to $T_p \rightarrow X$: it is the restriction of the external tensor product on the fibre product [Stacks, Tag 0DNK, Tag 01PU]. The resolution $U_p \rightarrow T_p$ has an invariant relatively ample divisor A_p by the preceding blowup construction. Hence

$$(28) \quad L_p = \mathcal{O}_{U_p}(A_p) \otimes \sigma_p^* B_p^{\otimes N_p}, \quad N_p \gg 0,$$

is g_p -invariant and η_p -ample; here σ_p^* uses the factorization through T_p . On the disjoint union (7), define M_p by

$$(j_\sigma^{(p)})^* M_p = L_q \quad (\sigma : [p] \rightarrow [q]).$$

Then $f_p^* M_p \simeq M_p$ and M_p is π_p -ample. Choose D_p with $\mathcal{O}_{X_p}(D_p) \simeq M_p$, componentwise [H77, II, Proposition 6.15]. This proves (21)–(22); the ample twist (23) follows from [Stacks, Tag 0892].

Computing the cohomology. Proper cohomological descent gives (16) [D74, (5.3.5), (5.3.8), and (8.2.1)]. More explicitly, put $C^{p,q} = C_{\text{sing}}^q(X_p^{\text{an}}, \mathbb{Q})$, with horizontal differential the alternating sum of face pullbacks and vertical differential d_{sing} . Then

$$\text{Tot}^k C^{\bullet,\bullet} = \bigoplus_{p+q=k} C^{p,q}, \quad D|_{C^{p,q}} = \delta + (-1)^p d_{\text{sing}},$$

and the augmentation induces a quasi-isomorphism from the singular cochain complex of X^{an} to this total complex. Filtering by simplicial degree gives (17) and (18). The simplicial identities imply $\delta^2 = 0$. Equivariance follows from the identities in part (1). Deligne's construction makes this spectral sequence compatible with mixed Hodge structures [D74, Proposition (8.3.5)]; this statement uses the mixed Hodge complexes underlying the construction, not a Hodge filtration on the raw singular cochains.

Purity, degeneration, and weights. Since X_p is smooth projective, $H^q(X_p, \mathbb{Q})$ is pure of weight q . The maps $\delta^{p,q}$ are morphisms of Hodge structures, so their cohomology $E_2^{p,q}$ is also pure of weight q . For $r \geq 2$,

$$d_r : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}$$

is zero: inductively, its source and target are pure of distinct weights q and $q - r + 1$. This proves $E_2 = E_\infty$. The induced filtration is Deligne's weight filtration with the indexing in (19), by [D74, Proposition (8.1.20)(ii) and §8.2]. The first-quadrant condition gives the stated weight range. The edge map is π_0^* and identifies $H^k(X, \mathbb{Q})/W_{k-1}$ with $E_2^{0,k} = \ker \delta^{0,k} \subset H^k(X_0, \mathbb{Q})$; this proves (20) (compare [D74, Proposition (8.2.5)]).

The semisimplicity consequence. Invariant subspaces and quotients of a semisimple operator are semisimple. The three formulas (24)–(26) therefore give semisimplicity on every weight-graded piece under the stated hypotheses; the action on $H^0(X_2, \mathbb{Q})$ has finite order. To pass from these pieces to $H = H^2(X, \mathbb{Q})$, set $T = f^*$ and let $P \in \mathbb{Q}[t]$ be the least common multiple of their minimal polynomials (omitting zero pieces). Then P is square-free and $S = P(T)$ is an endomorphism of mixed Hodge structures with $S(W_w H) \subseteq W_{w-1} H$. Strictness [D71, Theorem 2.3.5] gives

$$S(W_w H) = \text{im } S \cap W_w H.$$

Starting from $W_2 H = H$, repeated use of this identity forces $\text{im } S \subseteq W_1 H, W_0 H, W_{-1} H = 0$ successively. Hence $P(T) = 0$, proving semisimplicity on H . $\square$

Remark 1 (Translation into Hartshorne's varieties and divisors). Write

$$X_p = \coprod_{\alpha \in I_p} X_{p,\alpha}, \quad D_p = (D_{p,\alpha})_{\alpha \in I_p}.$$

Each $X_{p,\alpha}$ is a nonsingular projective variety in the classical sense. Since it is locally factorial, its Cartier divisors can be written as Weil divisors

$$D_{p,\alpha} = \sum_j n_{\alpha j} Z_{\alpha j}, \quad n_{\alpha j} \in \mathbb{Z},$$

where $Z_{\alpha j}$ are irreducible codimension-one subvarieties. Cartier and Weil divisors agree, and divisor classes modulo linear equivalence correspond to invertible sheaves [H77, II, Propositions 6.11, 6.15 and Corollary 6.16].

Fix $\ell \geq 0$. Replace f and all f_p by their N -th iterates, where $N \geq 1$ is a common multiple of the orders of the component permutations for $0 \leq p \leq \ell$. Each f_p in this range now fixes every component, and (21) remains valid. Write

$$f_{p,\alpha} := f_p|_{X_{p,\alpha}} : X_{p,\alpha} \xrightarrow{\sim} X_{p,\alpha}.$$

Then $f_p^* D_p \sim D_p$ means precisely that, for each $\alpha \in I_p$, there exists $\varphi_{p,\alpha} \in \mathbb{C}(X_{p,\alpha})^*$ such that

$$(29) \quad f_{p,\alpha}^* D_{p,\alpha} - D_{p,\alpha} = \operatorname{div}(\varphi_{p,\alpha}).$$

For (24)–(26), take $\ell = 3$. Positive iterates preserve semisimplicity of invertible operators over $\mathbb{Q}$.

The second assertion in (21) says that, for an ample Cartier divisor H on X ,

$$[D_{p,\alpha}] + m(\pi_p|_{X_{p,\alpha}})^*[H]$$

is an ample divisor class for $m \gg 0$.

On the possibly singular base X , one uses *Cartier* divisors; an arbitrary Weil divisor need not define an invertible sheaf. A component map $X_{p,\alpha} \rightarrow X$ need not be dominant. Accordingly, $\pi_p^*[H]$ denotes the class of $\pi_p^*\mathcal{O}_X(H)$: pulling back a chosen divisor by its local equations requires those equations to remain nonzero divisors. Pullback of its line bundle always exists; compare [Stacks, Tag 0C4S].

Remark 2 (Scope of the construction). The integer p is a simplicial degree, not $\dim X_p$; different components can have different dimensions. The theorem chooses the hypercover and its divisors together. Invariance concerns linear equivalence at each level, and no compatibility of the D_p with face or degeneracy pullbacks is asserted. The bundles denoted by M_p in applications are exactly $\mathcal{O}_{X_p}(D_p)$; in particular, $f_p^*c_1(M_p) = c_1(M_p)$. The auxiliary ample divisor H on X need not be f -invariant.

REFERENCES

- [BM08] E. Bierstone and P. D. Milman, *Functoriality in resolution of singularities*, Publ. Res. Inst. Math. Sci. **44** (2008), 609–639. Theorem 1.1. [Publisher PDF](#).
- [D71] P. Deligne, *Théorie de Hodge. II*, Publ. Math. Inst. Hautes Études Sci. **40** (1971), 5–57. Theorem 2.3.5 (strictness of morphisms of mixed Hodge structures). [Numdam PDF](#).
- [D74] P. Deligne, *Théorie de Hodge. III*, Publ. Math. Inst. Hautes Études Sci. **44** (1974), 5–77. Sections 5.3 and 6.2 (descent and hypercovers); Propositions (8.1.20), (8.2.2), (8.2.5), and (8.3.5) (cohomology, mixed Hodge structures, and weights). [Numdam PDF](#).
- [H77] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics **52**, Springer, 1977. Chapter II, §6, especially Propositions 6.11, 6.15 and Corollary 6.16. [Publisher page](#).
- [Stacks] The Stacks Project Authors, *The Stacks Project*. [Tag 0AMA](#) (matching objects); [Tag 01VG](#) (relative ampleness); [Tag 02OS](#), [Tag 02NS](#), [Tag 0809](#) (blowups); [Tag 0C4K](#) (composition of relatively ample sheaves); [Tag 0DNK](#), [Tag 01PU](#) (products and closed restrictions); [Tag 0892](#) (an ample twist from the base); [Tag 0C4S](#) (divisors and invertible sheaves). Tags accessed September 6, 2026.